Task Sheet Week 9 Day 2

Task 1: Read the provided introduction section and answer:

I. INTRODUCTION

T HE retina is one of the main components of the eye, which supports the visual function. It is located at the back of the eye, and its main job is to transform the light that enters through the eye to electrical signals that are passed on to the brain through the optical nerve. Due to its nature, the retina can both manifest the occurrence of diseases limited to the eyes, as well as broader scope physiological conditions, specifically, circulatory and brain diseases [1].

Diseases, such as age-related macular degeneration (ARMD), diabetic retinopathy (DR), and glaucoma cause blindness to more than 10 million people around the world every year [1]. Indeed, glaucoma is the second most common cause of blindness in the developed world [2], with ARMD being the most common cause of blindness for people above 50 years old [2], and DR is one of the most important causes of vision loss for people in the age group from 25 to 74 years [1].

Regular examination of the retina may support the early diagnosis of diseases before the occurrence of any symptoms. Early diagnosis is crucial since early detection may prevent total vision loss in patients and support delaying and potentially stopping degenerative diseases, e.g., progressive retinal atrophy, through a timely treatment regime.

Automated analysis and diagnosis systems have a significant impact in medicine and biology [3]. Computer-aided analysis (CAD) of retinal images can, for instance, help physicians in disease diagnosis, and early treatment planning, reduce the time taken to process large datasets and minimize variability in image interpretation [1]. Moreover, automatic analysis offers several advantages over manual inspection, being more cost-effective, objective, reliable, and relaxing the requirement for trained specialists to grade images [4]. On the other hand, manual inspection tends to be mundane, time-consuming, and requires proficient skills [5]. Indeed, one of the major stumbling blocks for manual retinal examination in developing countries is the lack of a sufficient number of qualified medical personnel per capita to diagnose diseases [6]. Prior to the development of deep learning methods, CAD systems were applied in various stages of the retinal diagnostic procedure, including image enhancement and restoration. Some CAD approaches attempted to imitate the means that clinicians perform retinal disease diagnosis, for instance, by performing image segmentation, feature extraction, and finally using machine learning [7]. For example, in [8], a total of 32 local binary patterns (LBP) multi-scale texture features per image were used as an input to various classifiers, including instance-based multi-label learning model, Multi-label Support Vector Machine Learning, Multi-label Learning neural network Radial Basis Function and Back-Propagation Multi-label Learning with promising performance. In summary, state-of-the-art attempts at tackling multi-label retinal diagnosis relied on the use of feature engineering, coupled with traditional machine learning algorithms, which, however, have substantial limitations, in regard to the suitability and distinguishability of features in the context of multiple simultaneous retinal diseases.

One of the most successful approaches to the automatic detection of retinal diseases is the use of Deep Learning (DL) techniques, specifically, Convolutional Neural Networks (CNN) and more recently, Transformer architectures. A considerable amount of work has been carried out on detecting the presence of common retinal diseases, such as ARMD, DR, Glaucoma, etc. For instance, Zago et al. [9] developed a system that uses two CNNs (pre-trained VGG16 and CNN) to diagnose DR according to the probability of lesion patches. Jiang et al. [10] developed a system based on three CNNs, namely, Inception-v3, ResNet152, and Inception-ResNet-v2, to classify fundus images into referable DR or non-referable DR. Burlina et al. [9] applied deep learning in detection and classification of ARMD using two deep convolutional networks, one was trained for the detection of ARMD, while the other used transfer learning.

Despite the promising results obtained in the detection of a single disease, the associated models are not sufficiently flexible to accommodate the simultaneous presence of multiple retinal diseases, which is commonly the case in real-world applications. Instead, clinicians require a diagnostic tool capable of detecting a diversity of conditions, simultaneously affecting a patient to provide the best possible treatment regime. Therefore, contrary to the vast majority of state-of-the-art works, which focus on detecting a single retinal disease, a higher value solution is multi- label disease classification, which would support simultaneous detection of a wide range of conditions present in a patient.

There are various challenges when dealing with multi-label classification in fundus imaging. For example, there is variability in the spatial extent of diseases, where some localize in specific regions of the retina, e.g., glaucoma, whereas others manifest themselves all over the retina (e.g., tessellation (TSLN) [11]). Simultaneous diagnosis of such diseases thus requires powerful architectures, capable of detecting changes in both small and large spatial regions of the retina.

Another issue is the scarcity of data. Most of the existing datasets are affected by different problems that make them unsuitable to satisfactorily train a multi-label model. For instance, they may focus on single diseases, contain a small number of samples, or indeed, a small number of pathologies to predict. Solving this problem requires the combination of existing datasets, to create an appropriate image database that can be used in model development.

Finally, a common problem when working with multi-label datasets is class imbalance. Usually, a small number of disease labels contain most samples, whereas most labels have only a few images. Thus, techniques designed to deal with the class imbalance problem are required, either by suitably modifying the dataset or the means that the model learns from the data.

In this work, a novel pipeline for multi-label disease classification based on fundus images is proposed.

A new dataset that combines publicly available fundus datasets for both single and multi-disease detection is generated to address the scarcity of publicly available data and to alleviate the high-class imbalance present in publicly available datasets. Preprocessing is applied to preserve the quality of the data and generate a final retinal image dataset that contains a wide variety of diseases to predict with enough samples per disease label.

Finally, a transformer-based model optimized through extensive experimentation is employed for multiple retinal disease classification, using the proposed dataset. The model is trained using a novel scheme, optimized through a series of experiments on its architectural design and hyperparameters, and by using a variety of techniques to partly alleviate the effect of the class imbalance present in the MuReD dataset over the model performance.

The main contributions of this work can be summarized as follows:

1) A new customized multi-label dataset, the MuReD dataset, is generated, which contains 20 disease classes, gathered from state-of-the-art sources and cleaned using an automatic quality score based on the sharpness and brightness of the image.

2) A transformer-based model optimized through extensive experimentation is used for the first time to detect and classify multiple retinal diseases.

1. **What is the main domain of the study?**

2. **What problem is being discussed?**

3. **Why is this problem important?**

4. **What existing solutions are mentioned? (List at least 3)**

5. **What limitations are identified?**

6. **What is the research gap?**

7. **What solution is proposed?**

Task 2: Expand your problem statement (previous task) into introduction (300 words) by incorporating new elements learned in today's lecture. Use the given structure:

Paragraph 1: Background + Problem

- Introduce the domain
- Describe the problem
- Show importance

Paragraph 2: Existing Work + Limitations

- Briefly mention existing approaches (use IEEE citation)
- Identify their limitations

Paragraph 3: Gap + Proposed Solution

- Clearly state the research gap
- Present your proposed solution
- Add contributions or objectives